CS410 Midterm 1 Part (a)

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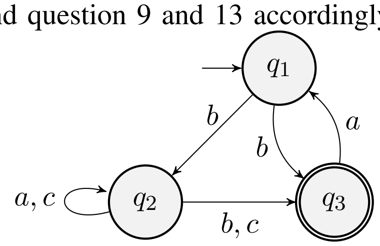
I. Multiple Choice Questions (40 points)

Given the following NFA M_1. Answer questions 1-5 and question 9 and 13 accordingly.



1. What is Q?

a) q_1
b) $\{q_1\}$
c) q_1, q_2, q_3
d) $\{q_1, q_2, q_3\}$
e) None

2. Which of the following can be the input alphabet Σ ?

a) {a, b}
b) {b, c}
c) {a, b, c}
d) {a, b, c, ϵ}
e) None

3. What is F?

a) $\{q_1, q_2\}$
b) q_3
c) $\{q_3\}$
d) q_1, q_2
e) None

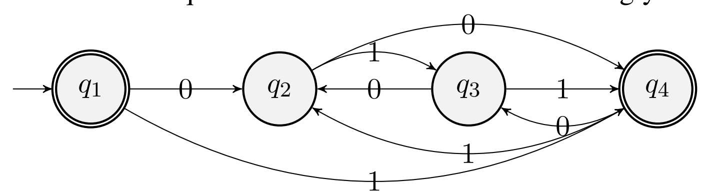
4. What is δ(q_2, c)?

a) q_2
b) $\{q_2\}$
c) q_2, q_3
d) $\{q_2$, $q_3\}$
e) None

5. Which of the following string is accepted by M_1?

a) ϵ
b) bba
c) ba
d) cab
e) None

Given the following DFA M_2. Answer questions 6-7 accordingly.



6. Which of the following change does not violate the DFA properties when applied alone?

a) Adding an ϵ transition from q_1 to q_2
b) Adding q_2 as another start state.
c) Adding q_3 as another final state.
d) Adding a 0 transition from q_2 to q_3.
e) Removing the 1 transition from q_3 to q_4.

7. Which states will be traversed for the input 101110?

a) $q_1q_4q_3q_4q_2q_3q_2$
b) $q_1q_2q_3q_4q_2q_3q_2$
c) $q_1q_4q_2q_4q_2q_3q_2$
d) $q_1q_4q_3q_2q_2q_3q_2$
e) $q_1q_4q_3q_4q_3q_3q_2$

8. How many states do we need at **minimum** if we want to design a DFA for the following language $L = \{w|w$ contains 1010 as a substring$\}$?

a) 3
b) 4
c) 5
d) 8
e) 16

9. If the NFA M_1 in Question 1 is converted to a DFA, how many states and how many final states will it have?

a) 4, 2
b) 5, 3
c) 6, 2
d) 7, 4
e) 8, 4

10. How many states do we need at **minimum** if we want to design an NFA for the following language $L = \{w|w$ contains a multiple of 4 a's or a multiple of 3 a's$\}$?

a) 4
b) 6
c) 8
d) 10
e) 12

11. If A and B are regular languages with n and m words ($n \leq m$), what is the **minimum** number of words for the language $L_1 = A \cup B$, $L_2 = A \cap B$?

a) m, n
b) m, 0
c) m + n, 0
d) m + n, m
e) m + n, n

12. Given the languages A = $\{00, 11\}$ and B = $\{00, 10\}$, which of the following is correct?

a) $A \cup B = \{00, 11, 10, 01\}$
b) $A \cap B = \{11\}$
c) $A^* = \{00, 11, 0011, 1100, 0000, 1111, \ldots\}$
d) $A \circ A = \{0000, 0011, 1100, 1111\}$
e) $B \circ B = \{0000, 1010, 1000, 0001\}$

13. What will be the corresponding regular expression between node q_1 and node q_3, after the node q_2 is removed from the given NFA M_1 in Question 1?

a) $b(a \cup c)(b \cup c)$
b) $b(a \cup c)^*(b \cup c)$
c) $(b(a \cup c)^*(b \cup c)) \cup b$
d) $(b(a \cup c)(b \cup c)) \cup b$
e) None of the above

14. Which of the following language is defined with the regular expression $\Sigma^*0\Sigma^*1\Sigma^* \cup \Sigma^*1\Sigma^*0\Sigma^*$?

a) $A = \{w|w$ contains a 1 after each 0$\}$
b) $B = \{w|w$ contains at least one 0 and one 1$\}$
c) $C = \{w|w$ contains the string 01 as a substring$\}$
d) $D = \{w|w$ starts and ends with 01$\}$
e) $E = \{w|w$ starts with 01$\}$

15. Which of the following is the regular expression for the language $A = \{w|$ length of w is a multiple of 3$\}$?

a) Σ^*
b) $\Sigma^*\Sigma^*$
c) $\Sigma^*\Sigma^*\Sigma^*$
d) $\Sigma(\Sigma\Sigma)^*$
e) $(\Sigma\Sigma\Sigma)^*$

16. Given the language $A = \{1^n0^n|n \geq 0\}$, and p the pumping length. Let say we wanted to prove that A is not regular. Let say the selected string is 1^p0^p. Which of the following can be string y, that can be pumped?

a) 0^{p+1}
b) 1^p
c) 1100
d) 0000
e) 10

17. Given the language $A = \{0^i1^j|i > j\}$, and p the pumping length. Let say we wanted to prove that A is not regular. Which of the following string would you choose to prove it?

a) $0^{2p}1^{2p}$
b) 0^p1^p
c) $0^{p+1}1^p$
d) $0^{p/4}1^{p/4}$
e) 01

18. Which of the following languages is not regular?

a) $A = \{a^nb^{2n}|n \geq 0\}$
b) $B = \{a^{2n}b^mc|n, m \geq 0\}$
c) $C = \{a^{3n+2}b^{m+1}|m, n \geq 0\}$
d) $D = \{a^kb^mc^n|k \geq 2, m \geq 0, n \leq 2\}$
e) All of them are regular

19. Which of the following languages is regular?

a) $A = \{0^n1^n|n \geq 0\}$
b) $B = \{w|w$ has an equal number of 0s and 1s$\}$
c) $C = \{w|w$ has an equal number of occurrences of 01 and 10 as substrings$\}$
d) $D = \{ww|w \in \{0, 1\}^*\}$
e) None of them is regular

20. Which of the following languages is not regular?

a) $A = \{1^n|n \geq 0\}$
b) $B = \{1^n|n$ is prime$\}$
c) $C = \{1^n|n$ is odd$\}$
d) $D = \{1^n|n$ is even$\}$
e) All of them are regular